

Accurate & real-time food classification through the synergistic integration of EfficientNetB7, CBAM, transfer learning, and data augmentation

Shayan Rokhva , Babak Teimourpour *

Department of Information Technology Engineering, Faculty of Industrial and Systems Engineering, Tarbiat Modares University, Tehran, Iran

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ABSTRACT

The integration of artificial intelligence into modern society has become a powerful force, significantly enhancing efficiency by streamlining various daily activities. In the food industry, AI-driven recognition systems provide valuable benefits, such as improved nutrient tracking, reduced food waste, and increased productivity in food production and consumption settings. Central to the success of these systems is accurate food classification, a critical process that directly influences downstream operations. Despite advances in this field, there is still a need for models that improve accuracy without compromising processing speed, prompting researchers to develop cost-effective, high-precision solutions. This study addresses this gap by introducing a model that leverages the state-of-the-art pre-trained EfficientNetB7 architecture for effective feature extraction, further enhanced through data augmentation, fine-tuning, hyperparameter optimization, and the integration of a Convolutional Block Attention Module (CBAM), which improves both channel and spatial attention. We trained, validated, and evaluated the model on the Kaggle Food11 dataset, comprising 16643 images across 11 imbalanced classes with substantial intra-class variations and inter-class similarities. The proposed model achieved an impressive average accuracy of 96.40 %, surpassing all prior studies utilizing single deep learning models, and closely approached the highest reported accuracy of 96.88 %, achieved by an ensemble method of four models. Remarkably, the proposed model could classify approximately 62 unseen data per second during inference mode, highlighting its applicability for real-time food classification and efficiency improvements in subsequent processes. Finally, a detailed discussion provides valuable insights into the model's effectiveness and robustness, the study's challenges, and recommendations for future enhancement.

1. Introduction

1.1. The concept and importance of study

The classification, detection, and segmentation of food images using AI and computer vision offer significant potential for innovation and efficiency. These technologies facilitate and automate nutritional tracking, support AI-powered kitchens, and address food waste management (Lubura et al., 2022; Rokhva et al., 2024). Additionally, with slight modifications, food image recognition techniques can be adapted to monitor crops and other items throughout the supply chain, boosting productivity across food production and consumption processes (Attri et al., 2023; Zhou and Chen, 2023).

Developing accurate and robust models that run promptly on cost-effective hardware is crucial to ensure both effectiveness and efficiency. This has led researchers to explore diverse AI models for better

performance, generalization, and speed (Fang et al., 2023; Moumane et al., 2023). In many image analyses and their applications, classification is a critical initial step, as subsequent analyses depend on the accuracy and speed of recognition. Therefore, a model with high accuracy, robustness, and rapid processing of real-world extensive data is pivotal in improving food classification and overall efficiency in the food industry (Chakraborty and Aithal, 2024; Lubura et al., 2022).

Deep learning (DL), a branch of AI, is particularly effective for solving complex problems, especially working with large datasets. Convolutional Neural Networks (CNNs), combining DL with computer vision, are widely applied in visual recognition tasks. Their performance tends to improve as dataset sizes increase, with deeper and more complex models requiring more data (Alzubaidi et al., 2021).

While CNN architectures vary based on their specific use, they consistently follow a structured process involving feature extraction, information summarization, and image classification. CNNs use multiple

* Corresponding author.

E-mail addresses: Shayan_rokhva@modares.ac.ir (S. Rokhva), B.teimourpour@modares.ac.ir (B. Teimourpour).

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convolutional layers to extract and combine image features, creating a feature map that aids classification. Max-pooling layers, placed between the convolutional layers, perform feature selection and reduce map size, lowering computational costs. Average pooling, usually at the end, summarizes the most influential and enriched features extracted by previous processes. The final small, data-rich feature map is flattened and passed to the classifiers for prediction (Alzubaidi et al., 2021; Sarraf et al., 2021). Conversely, Artificial Neural Networks (ANNs) require exponentially more parameters for image classification, demanding significant computational resources, making them less practical for such tasks compared to CNNs (Lubura et al., 2022).

Over the past decades, various CNNs have been developed and implemented, including the AlexNet, VGG, GoogleNet, ResNet, MobileNet, and EfficientNet families. Each family contains models that differ in depth, size, and parameters while maintaining a similar structure and underlying concept. However, models among different families also vary in their overall architecture according to specific applications and concepts (Karypidis et al., 2022; Li et al., 2021). For instance, the MobileNet family is designed for quick classification with fewer parameters and computations, making them ideal for mobile and embedded devices. However, they may exhibit lower performance in specific applications (Banoth and Murthy, 2024; Rokhva et al., 2024). Conversely, the EfficientNet family, developed by Google, employs compound scaling to optimize depth and width, yielding lightweight models for more straightforward and less complex tasks and deeper models for complex applications requiring high accuracy. The enhanced performance of the deeper EfficientNet models has been demonstrated across numerous tasks (Huang et al., 2019; Tan, 2019).

The widespread adoption of these models emphasizes the need to utilize, customize, and improve them for real-world applications while tackling their challenges, such as inferior speed or accuracy, by leveraging DL techniques. (Alzubaidi et al., 2021; Karypidis et al., 2022). For example, while deep and dense models offer higher accuracy using a large number of data, they also lead to more trainable parameters and increased processing times (Tan, 2019). Therefore, leveraging pre-trained knowledge, freezing part of the network, utilizing parallel computation of GPU, and similar techniques can help address this issue (Farahani et al., 2021).

Since we briefly understood the challenges and importance of integrating AI and computer vision into kitchen and food consumption settings, we should also explore their frequent use in modern facilities (Chakraborty and Aithal, 2024). Smart refrigerators utilize hidden cameras and AI models to identify food items and notify users about freshness and expiration dates, thereby aiding in reducing food waste, which constitutes 25–35 % of food globally. However, these advanced facilities remain primarily unaffordable for the average consumer, probably due to the expenses of state-of-the-art research, highlighting a significant area for enhancement (Gao et al., 2019; Shweta, 2017).

To enhance accessibility for people from diverse backgrounds, research should prioritize optimizing performance and increasing speed while reducing dependence on expensive high-end graphical processing units (GPUs). Additionally, refining the performance and robustness of these systems is crucial (Alijani et al., 2024). Therefore, further exploration and advancements in this field are both necessary and warranted to enhance usability and efficiency.

1.2. Brief survey of similar works

This section aims to expand readers' understanding by exploring studies that utilize computer vision and DL techniques in the food industry and related fields. One of the early studies in automatic AI-driven food recognition (Kagaya et al., 2014) employed a CNN architecture on a large food image dataset, emphasizing hyperparameter tuning. Their findings demonstrated that CNNs outperformed Support Vector Machine (SVM), a traditional ML algorithm, using handcrafted features. Similarly, (Kawano and Yanai, 2014) reported that combining features from

Deep CNNs with traditional handcrafted image features, such as Fisher Vectors, HoG, and Color patches, led to enhanced food recognition performance, achieving top and top-5 % and top-1 % accuracies of 72.26 % and 92 %, respectively, on the UEC-FOOD100 dataset.

(Singla et al., 2016) utilized a Google-Net-based model, achieving impressive accuracies of 99.2 % for food and non-food classification and 83.6 % for differentiating food categories. (Siddiqi, 2019) applied the VGG16 model, renowned for its numerous trainable parameters, in conjunction with transfer learning, attaining a classification accuracy of 99.27 % for AI-driven fruit image recognition. Additionally, (Zhu et al., 2020) introduced a system capable of classifying and detecting multiple food items in refrigerators faster and more accurately than the previous method, boosting the previous F-measure baseline by 3–5 %.

In agriculture, a field close to food recognition, researchers (Kumar et al., 2020) showed that ResNet34 classifies crop images with an impressive accuracy of 99.4 %. A recent study (Banoth and Murthy, 2024) used MobileNetV2 for highly accurate soil image classification, exceptionally classifying test data with 100 % accuracy. These DL models have also been widely used for medical purposes. For example, EfficientNetV2 was applied for chest X-ray and CT image classification, achieving accuracy of 98.33 % and 97.48 %, respectively (Huang and Liao, 2022). For food classification, researchers (Fakhrou et al., 2021) utilized DCNNs with transfer learning (TL) and data augmentation, obtaining an accuracy of 95.55 %.

A study (Mazlounian et al., 2020) investigated the classification and segmentation of food waste using DL, TL, and data augmentation, achieving an accuracy of 83.5 % for classifying food waste images and a pixel accuracy of 98.5 % for segmentation. Another study in the food industry (Lubura et al., 2022) implemented a classic CNN model similar to the VGG architecture for food classification, achieving 98.8 % performance. This research also estimated food waste at 21.3 % by segmenting images from Serbian students' meals over six months in 2022.

Recent advancements in computer vision, such as attention mechanisms, have also been used to recognize food items. Nonetheless, their applications can be improved. In one research in the food industry (Hui-jiang et al., 2024) a CBAM-InceptionV3 model, embedding 11 CBAM modules into Inception V3, was proposed to enhance food image classification. Utilizing TL on the Food-101 dataset, the proposed model achieved 82.01 % accuracy, outperforming the original InceptionV3 in feature extraction, classification, and training efficiency.

Importantly, studies focusing on the underexplored Food11 dataset will be thoroughly analyzed and compared in the discussion section, addressing methodologies, results, and key considerations. Therefore, to prevent redundancy, they are not included here.

1.3. Research efforts

Given the crucial role of automated and precise food recognition, this study aims to advance multiclass food classification performance using the challenging Food11 dataset, comprising 16,643 images across 11 diverse classes. This dataset serves as an ideal test case due to its inherent challenges, including data imbalance, intra-class diversity, and inter-class similarities. In line with previous studies, the focus of this work is to improve model performance while also ensuring that inference speed remains sufficiently high for practical applications, an aspect often underappreciated in earlier research.

Through a comprehensive review of DL models and modules, we proposed the utilization of EfficientNetB7, a state-of-the-art and intricate backbone for feature extraction. To further enhance the feature map, the Convolutional Block Attention Module (CBAM) was incorporated, which improves both channel and spatial attention for more precise and refined feature representation. Fine-tuning, along with effective hyperparameter tuning, such as regularization, a decaying learning rate, and accelerators, was employed to facilitate faster and more stable convergence. Additionally, data augmentation techniques such as random flipping, rotation, color variation, and random erasing

were deliberately applied to prevent overfitting, bolster the model's robustness, and replicate real-world conditions where images may vary in perspective and lighting.

While the theoretical contributions of this study are not groundbreaking and align with established DL methodologies, its practical achievements are evident in its superior performance and efficient inference speed. Specifically, the proposed model, employing a single well-organized framework, outperformed all studies utilizing a single model, coming very close to the best performance reported by a weighted ensemble learning approach that combined four models. Notably, the results of this study even outperformed the vote ensemble learning approach. Furthermore, the model's inference speed of 62 images per second, using 256×256 images, is deemed satisfactory for numerous real-world applications. A detailed discussion provided compares the results with similar studies on the same dataset, further highlighting the impact of the utilized techniques.

2. Material and methods

2.1. Overview of research workflow

This research follows a standard ML workflow, focusing on model training with a significant portion of the dataset, optimizing performance with the validation set, and safeguarding evaluation data only for final testing. TL, data augmentation, and optimizing hyper-parameters are integrated during training and validation but not test data (Al-Alshaikh et al., 2024).

2.2. Hardware configuration

Due to the computational demands of DL and the unstructured nature of image data, which require extensive computations, GPUs are indispensable. However, to make this research more accessible and cost-effective for real-world applications, we opted for the budget-friendly T4 GPU with 16 GB from Google Colab, balancing efficiency and affordability.

2.3. Distribution & challenges of data

The Food-11 dataset (Food-11 image dataset, n.d.), sourced from Kaggle, contains 16643 images spanning 11 food categories. Fig. 1 illustrates the distribution, revealing a notable imbalance that could hinder model performance, especially for underrepresented classes. Approximately 60 % of the images are allocated to the training set in each class, with the remainder almost evenly split between validation and evaluation subsets.

The dataset also presents classification challenges due to substantial intra-class diversity and inter-class similarity. Broad categories like dairy products, desserts, and fried food include visually distinct items, complicating accurate classification. Additionally, food categories such as bread, dairy products, desserts, eggs, and fried food share similar bright colors such as light brown, golden, white, and yellow, raising the potential for misclassification and even mutual misclassification, ultimately undermining overall performance.

Additionally, foods such as hamburgers and pizzas, typically classified as bread in the dataset, are prone to misclassification with fried foods. Moreover, some images feature multiple objects but are categorized based on the most prominent one.

These visual similarities and challenges are particularly evident in the first five classes illustrated in Fig. 1, from Bread to Fried Foods, heightening the potential for misclassification across categories. Among other classes, vegetable-fruit is just a broad class. Nonetheless, the color hues are generally different than other categories, potentially facilitating recognition.

2.4. Transforming & augmenting data

The dataset images varied in dimensions. Consequently, they were all resized to 256×256 pixels, a common size in computer vision projects that strikes a balance between resolution and speed. After converting them to numerical values, normalization was applied using the mean and standard deviation (STD) of ImageNet. To enhance performance without increasing the training data, augmentation techniques like rotation, horizontal flipping, marginal color jittering, and partial erasing were randomly applied to the training set. Consequently, some images remained unchanged, while one or more techniques modified

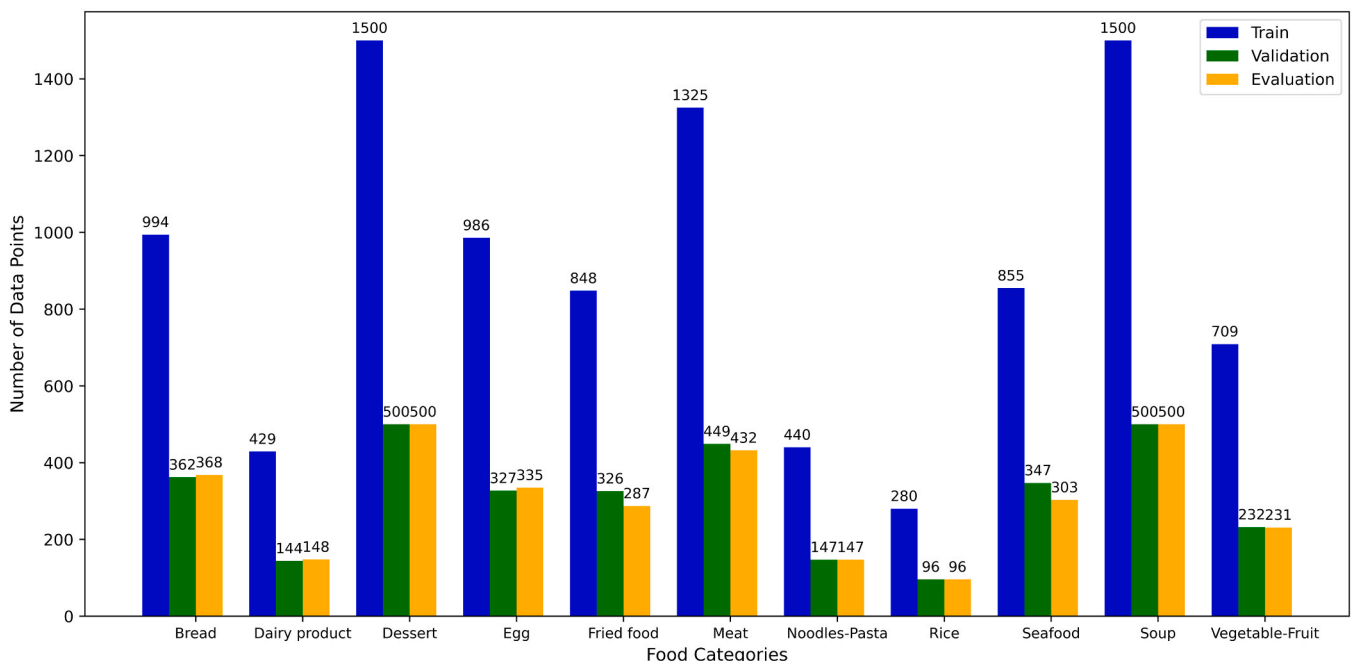


Fig. 1. Data distribution.

others. This approach increased the diversity of training data, generally leading to a more robust learning process and better model performance since the model was exposed to a broad range of data (Alijani et al., 2024).

2.5. EfficientNetB7 backbone

The EfficientNet family, introduced by Google in 2019 (Tan, 2019), constitutes a state-of-the-art architecture designed to enhance performance efficiently through a compound scaling approach. This methodology simultaneously scales the model's depth, width, and resolution, yielding superior performance compared to the isolated scaling of any single dimension.

Encompassing models from B0 to B7, the EfficientNet family includes lightweight variants such as B0, B1, and B2, which possess fewer trainable parameters and require less computational power and time (Tan and Le, 2021). However, these models may often exhibit lower performance on complex tasks. Conversely, the deeper models, such as B5, B6, and B7, are characterized by increased density and computational intensity, rendering them more suitable for large-scale applications requiring high accuracy (Tan, 2019).

Table 1 comprehensively compares EfficientNet models with other prominent computer vision architectures on the 1000-class ImageNet dataset, underscoring the superior performance of EfficientNets, with considerably fewer parameters and FLOPs. Such efficiency is attributed to the innovative compound scaling technique employed within this family of architectures, which optimally balances model complexity with performance (Arnandito and Sasongko, 2024; Tan, 2019).

EfficientNetB7, renowned for its exceptional performance in complex tasks (Tan, 2019), was selected as the backbone for feature extraction. As illustrated in Fig. 2, this architecture (Khalil et al., 2022) comprises Mobile Inverted Bottleneck Convolutions (MBConv) organized into seven blocks, beginning with a standard Conv2D layer and progressing through increasingly deeper and wider MBConv layers that extract progressively detailed features. The deeper blocks initially utilize broader 5×5 convolutions before transitioning to 3×3 convolutions, thereby capturing more abstract, high-level information.

Moreover, EfficientNetB7 employs depth-wise separable convolutions within its MBConv layers, significantly reducing computational complexity while preserving feature richness. Each MBConv block

consists of an expansion phase, where the input channels are increased to capture intricate details, followed by a compression phase that reduces dimensionality. As the network deepens, later blocks adopt larger kernel sizes and higher expansion ratios, effectively increasing the receptive field; this enhancement enables the model to summarize input data from even small objects more efficiently while extracting complex patterns and abstract representations.

This study sets the input image dimensions at $256 \times 256 \times 3$, representing a 256×256 image size across three channels: red, green, and blue (RGB). Consequently, the resulting feature map measures $8 \times 8 \times 2560$, showcasing the 8×8 -sized feature map enriched with summarized data and a large receptive field. With no modification, the final feature map in EfficientNetB7 is flattened and passed through a fully connected (FC) layer for classification; however, in this study, the feature map undergoes further optimization using the CBAM, which integrates channel and spatial attention mechanisms before reaching the FC layer.

2.6. Convolutional Block Attention Module (CBAM)

The Convolutional Block Attention Module (CBAM) is an innovative attention mechanism designed to enhance the performance of CNNs. It does this by refining feature maps through two fundamental processes: focusing on important channels and emphasizing significant spatial regions. CBAM aims to improve the model's ability to recognize and prioritize relevant image features, particularly useful in tasks like image classification and object detection. As CBAM's structure is shown in Fig. 3, it consists of a Channel Attention Module (CAM) and a Spatial Attention Module (SAM), operating sequentially, CAM followed by SAM (Woo et al., 2018).

CAM is responsible for evaluating the importance of different channels within the feature map through a three-step process. The first step involves information gathering, where CAM collects global data from the feature map by summarizing values across all spatial locations. As shown in Fig. 3, this is accomplished using two techniques: global average pooling, which averages the values, and global max pooling, which selects the maximum value. Consequently, these methods yield two distinct representations of channel importance. The second step focuses on weight calculation; here, the two summaries are processed through a small neural network, which learns to assign weights to each

Table 1

A comprehensive comparison of EfficientNets and other computer vision models.

Model	Top-1 Accuracy	Top-5 Accuracy	Parameters	Parameters/ EfficientNet	FLOPs	FLOPs/ EfficientNet
EfficientNet-B0	77.1 %	93.3 %	5.3 M	1x	0.39B	1x
ResNet-50 (He et al., 2016)	76.0 %	93.0 %	26 M	4.9x	4.1B	11x
DenseNet-169 (Huang et al., 2017)	76.2 %	93.2 %	14 M	2.6x	3.5B	8.9x
EfficientNet-B1	79.1 %	94.4 %	7.8 M	1x	0.7B	1x
ResNet-152 (He et al., 2016)	77.8 %	93.8 %	60 M	7.6x	11B	16x
DenseNet-264 (Huang et al., 2017)	77.9 %	93.9 %	34 M	4.3x	6.0B	8.6x
Inception-V3 (Szegedy et al., 2016)	78.8 %	94.4 %	24 M	3.0x	5.7B	8.1x
Xception (Chollet, 2017)	79.0 %	94.5 %	23 M	3.0x	8.4B	12x
EfficientNet-B2	80.1 %	94.9 %	9.2 M	1x	1.0B	1x
Inception-V4 (Szegedy et al., 2017)	80.0 %	95.0 %	48 M	5.2x	13B	13x
Inception-ResNet-V2 (Szegedy et al., 2017)	80.1 %	95.1 %	56 M	6.1x	13B	13x
EfficientNet-B3	81.6 %	95.7 %	12 M	1x	1.8B	1x
ResNeXt-101 (Xie et al., 2017)	80.9 %	95.6 %	84 M	7.0x	32B	18x
PolyNet (Zhang et al., 2017)	81.3 %	95.8 %	92 M	7.7x	35B	19x
EfficientNet-B4	82.9 %	96.4 %	19 M	1x	4.2B	1x
SENet (Hu et al., 2018)	82.7 %	96.2 %	146 M	7.7x	42B	10x
NASNet-A (Zoph et al., 2018)	82.7 %	96.2 %	89 M	4.7x	24B	5.7x
AmoebaNet-A (Real et al., 2019)	82.8 %	96.1 %	87 M	4.6x	23B	5.5x
PNASNet (Liu et al., 2018)	82.9 %	96.2 %	86 M	4.5x	23B	6.0x
EfficientNet-B5	83.6 %	96.7 %	30 M	1x	9.9B	1x
AmoebaNet-C (Cubuk et al., 2019)	83.5 %	96.5 %	155 M	5.2x	41B	4.1x
EfficientNet-B6	84.0 %	96.8 %	43 M	1x	19B	1x
EfficientNet-B7	84.3 %	97.0 %	66 M	1x	37B	1x
GPipe (Huang et al., 2019)	84.3 %	97.0 %	557 M	8.4x	-	-

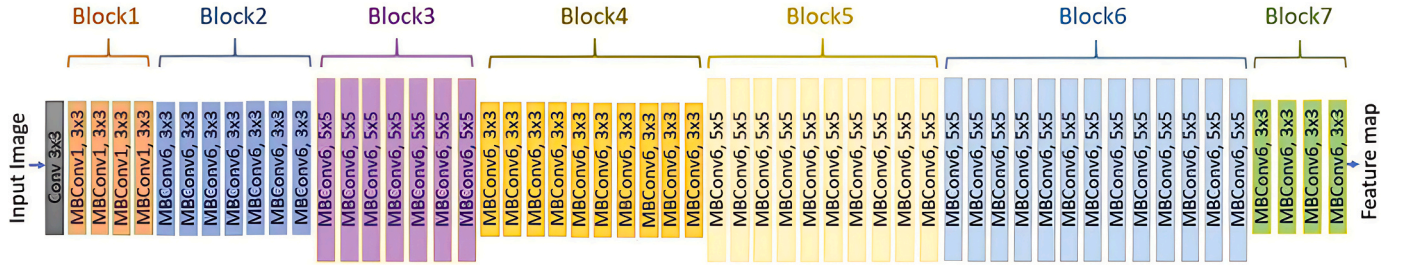


Fig. 2. EfficientNetB7 backbone structure.

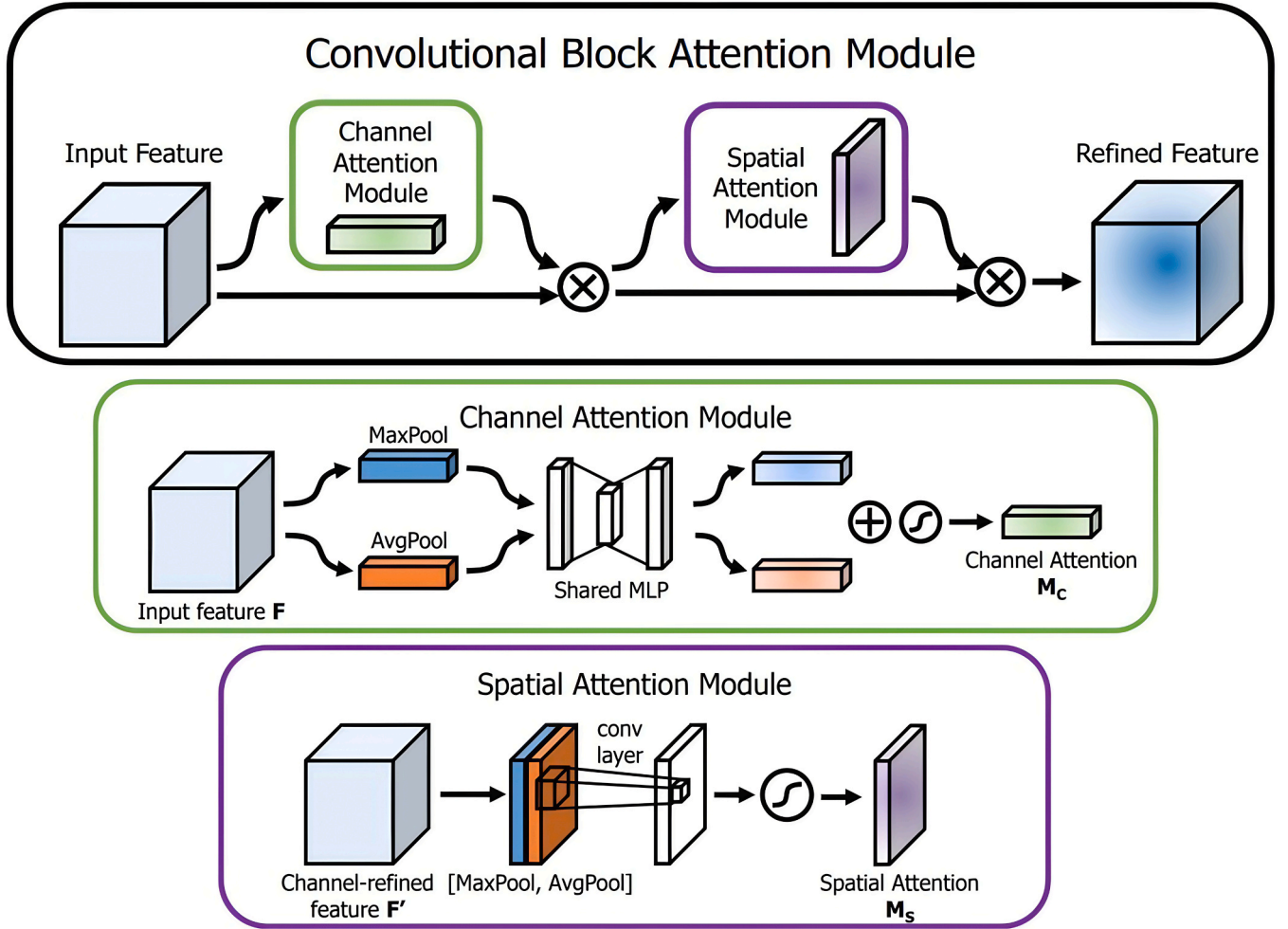


Fig. 3. Convolutional Block Attention Module (CBAM) and its components.

channel based on its relevance to the task. Finally, these calculated weights are applied back to the original feature map in the feature enhancement phase. Channels identified as more critical receive higher weights, amplifying their contribution, while less significant channels are diminished.

After refining channel-wise features through CAM, SAM focuses on enhancing specific spatial regions within those channels through a three-step process. As depicted in Fig. 3, the initial step involves extracting contextual information, where SAM analyzes the output from CAM to gather insights about spatial regions by applying both average pooling and max pooling across the channels. This dual approach helps identify which areas of the feature map hold greater significance. The second step is the creation of an attention map; here, the pooled information is combined and processed through a convolutional layer to

generate a spatial attention map that highlights critical regions in the feature map for further processing. Finally, in the refinement phase, this spatial attention map is applied to the refined feature map obtained from CAM. Areas identified as essential are emphasized, enabling the model to focus on crucial parts of the input image.

By combining these two processes, CBAM allows deep learning models to dynamically adjust their focus on features that matter most, leading to improved performance in various computer vision applications. This approach enriches feature representation and helps models become more interpretable and efficient in processing visual data. The efficient application of the CBAM module has been proven in numerous studies since its invention. In this study, the extracted feature map from EfficientNetB7, which is rich in data, is modified more through the application of CBAM. Afterward, it is flattened and goes through the FC

layer for classification.

2.7. Attention-based classification framework

The proposed framework, illustrated in Fig. 4, is an advanced method that can address numerous real-world applications, particularly those facing challenges such as intra-category diversity and inter-category similarity. Consequently, in line with previous research, focusing on improving the performance of food classifications, this model was selected and thoroughly analyzed.

The general diagram outlines how the image data, following pre-processing, is processed through EfficientNetB7, a single cutting-edge model specifically chosen for its superior feature extraction capabilities. As highlighted in Section 2.5, EfficientNet's advanced feature extraction stems from balancing depth, width, and resolution, rather than prioritizing any single aspect. The notable advantages of this compound scaling approach are demonstrated in Fig. 4 via heatmap (Tan, 2019).

The architecture of EfficientNetB7, as depicted, predominantly consists of MBConv6 blocks, utilizing both 3×3 and 5×5 kernels. These MBConv6 blocks expand the number of channels sixfold, enabling the extraction of richer and more detailed features. Following this expansion, the Squeeze-and-Excitation (SE) network enhances channel-wise attention, after which a 1×1 convolution is applied for the compression phase. Then, the initial feature maps are reintroduced through residual connections to retain crucial information from earlier stages.

Despite the large number of parameters in EfficientNetB7, the EfficientNet family, as noted in Table 1, demonstrates exceptional efficiency compared to similar models in terms of parameters and FLOPs. This efficiency is primarily attributed to the well-designed MBConv6 architecture, employing the expansion phase for enhanced feature extraction, the SE network for channel-wise attention, the compression phase for summarizing outputs and lowering computational overhead, and residual connections for retaining critical initial features that could

otherwise be lost within the block. While few MBConv blocks may struggle in highly complex tasks, similar to the limitations seen in MobileNet models for very intricate tasks, using a sufficient number of these blocks with proper structuring can achieve excellent performance with balanced parameters and computational costs, effectively optimizing both accuracy and efficiency, which is something required given the scope of this study.

According to Fig. 4, MBConv blocks employ either 3×3 or 5×5 kernels, each serving distinct purposes. The 3×3 kernels, with their smaller receptive fields, are ideal for capturing local and general features, are computationally less demanding, and are thus more prevalent in the initial layers. Conversely, 5×5 kernels, offering larger receptive fields, excel in deeper layers where they combine features to generate richer and more intricate representations, albeit with greater computational costs. In this task, these larger kernels are particularly advantageous for combining a blend of general features, which are common across various food categories, and create more complex and class-specific features, in turn enhancing classification performance. Additionally, the residual connections in each MBConv block play a critical role in preserving important information from earlier layers, ensuring these key details are not lost during the processing.

After extracting the feature map using EfficientNetB7, it undergoes further refinement through the CBAM module, which sequentially applies CAM and SAM. The architectural details of CAM and SAM are provided in Section 2.6 and illustrated in Fig. 3, respectively. In this regard, Fig. 4 elaborates on their contribution to the overall method. Using Max Pooling, Average Pooling, and an MLP, CAM assigns weights to the feature map's channels based on their importance, allowing the model to prioritize influential channels and reduce attention to less significant ones. Following CAM, SAM operates within each channel to identify spatially significant regions, enabling the model to focus on crucial parts of the image, such as the food itself, while ignoring irrelevant surroundings.

The feature map generated by CBAM is now refined in terms of both channel and spatial attention. It is then flattened and passed through a

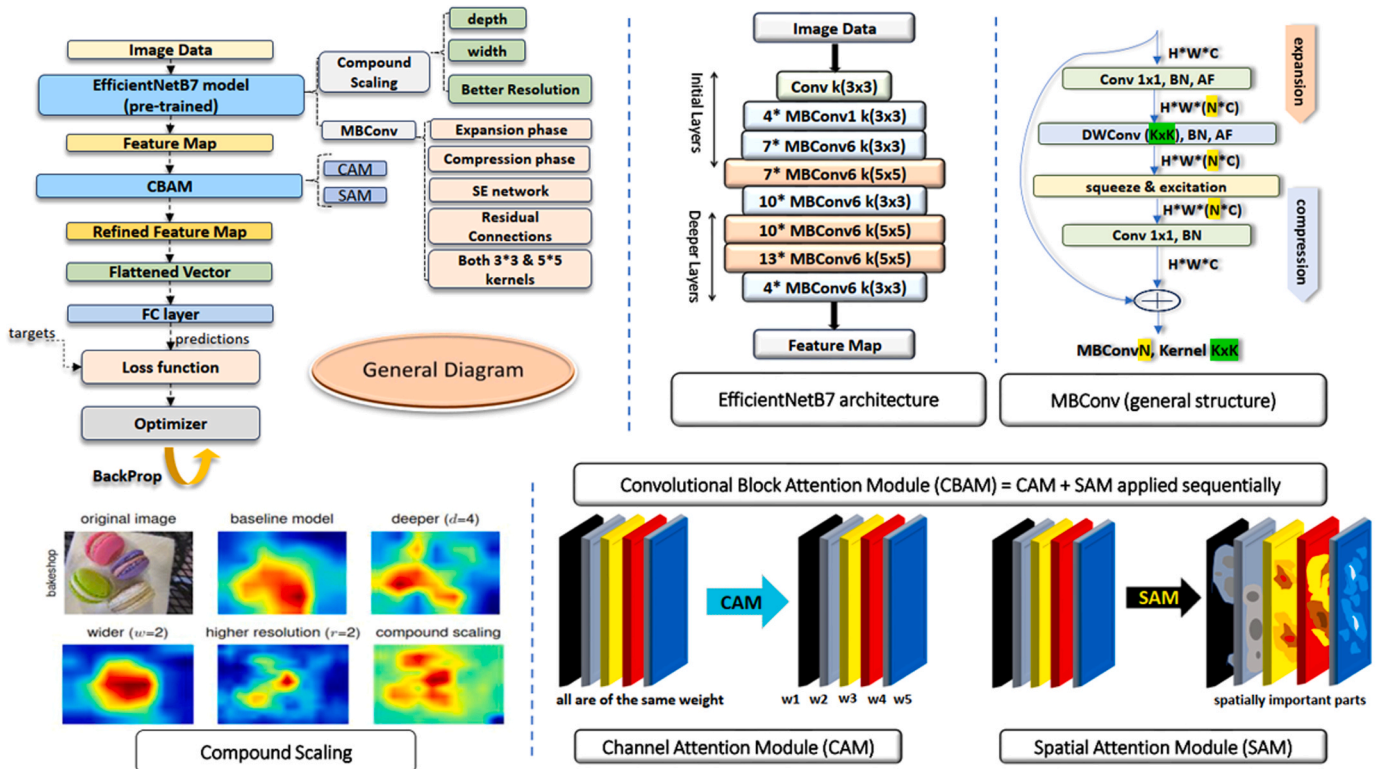


Fig. 4. The proposed attention-based classification framework.

FC layer containing 11 neurons, corresponding to the 11 classification classes. The model's predictions are compared with the actual targets using a loss function, and the optimizer adjusts the network's parameters via backpropagation. This process ensures that the pre-trained parameters of EfficientNetB7 and the CBAM module are all fine-tuned for optimal results. In Section 3, Results and Analysis, we will show that the proposed methodology achieves remarkable performance, outperforming many studies conducted on the same dataset.

2.8. Transfer Learning

Transfer Learning (TL) involves utilizing the weights and biases from a network trained on one task to enhance performance on a new application, effectively leveraging existing knowledge. This approach offers significant advantages, such as improving initial accuracy, achieving faster convergence, and reaching superior performance within the same number of epochs. In this context, the closer the two domains are, the more effective TL is expected (Hosna et al., 2022; Iman et al., 2023).

There are three main types of TL: Feature extraction, where all parameters are frozen except for those in the FC layer, making it suitable for scenarios with limited data or computational resources; Fine-tuning, which allows training of deeper layers responsible for extracting task-specific features while freezing earlier tasks with learning general features; and Full fine-tuning, where all parameters are retrained for additional epochs to adapt them specifically to the new task (Zhuang et al., 2020).

Research indicates that full fine-tuning typically yields better results when sufficient computational resources and large data volumes are available. Even though this method demands more computational power, once the model is optimized, its parameters can be utilized multiple times in inference mode without requiring backpropagation during training. Thus, intensive computation becomes a one-time necessity (Xing et al., 2024; Zhuang et al., 2020).

In this study, the EfficientNetB7 backbone, aimed at extracting rich and deep features, was initially imported with the weights and biases of extensive ImageNet data. Yet, to ensure optimal weights that are highly task-specific, all model parameters were exposed to more training epochs.

2.9. Loss function, optimizer & hyper-parameter tuning

Stochastic Gradient Descent (SGD) was employed as the optimizer, while Cross Entropy Loss served as the loss function. The learning rate (LR), the most influential parameter for model training, was set at 0.01. However, to achieve optimal performance, it was not fixed and halved every five epochs. This strategy allows for accelerated learning in the early stages of training, where faster convergence and larger steps are needed. Conversely, slower learning in later epochs facilitates a better and smoother convergence, leading to the global minimum loss.

L2 Regularization was set to $1e-5$ to prevent overfitting and enhance generalization. Momentum was configured at 0.9 to accelerate learning and convergence effectively, while the Nesterov accelerator was enabled for improved convergence. The batch sizes for training, validation, and evaluation were 16, 32, and 32, respectively. The smaller training batch size was chosen due to RAM constraints during backpropagation. It is crucial to report these batch sizes as they can marginally influence average processing speed; larger batch sizes can leverage GPU parallel computations if RAM permits, potentially slightly reducing total computation time.

2.10. Evaluating Performance & Speed

This study assesses performance using accuracy, a single number indicating the proportion of accurate predictions. Nonetheless, a confusion matrix was utilized to reveal errors' locations and portions,

highlighting classes prone to mutual misclassification. This graphical representation helps us extract insights, guide discussions, and identify improvement areas. Speed is also critical, varying slightly with batch sizes and differing heavily across GPUs. As mentioned, this study reports training and evaluation speeds using the affordable T4 GPU. Additionally, while both speeds are documented, evaluation speed is essential for real-world applications when the trained model is used in its inference mode.

2.11. Generalization & K-fold cross-validation

To enhance applicability, results should be generalized using K-fold cross-validation, a standard technique for improving robustness. However, this approach was not feasible due to the initial dataset separation. Therefore, to ensure robust results while avoiding data leakage or bias, the model was trained, optimized, and tested on the evaluation data five times, initializing all parameters from scratch each time. The mean performance from these runs was reported, yielding a generalized result that enhances robustness similarly to 5-fold cross-validation (Kohavi, 1995). In this context, we expect the five accuracies to be closely aligned but slightly varied due to the randomness in data loader creation and the application of data augmentation techniques on the training samples. The average accuracy should also align closely with each of the five reported accuracies for generalization.

3. Results & analysis

After multiple experiments, the predicted evaluation accuracies on unseen data were consistently close, aligning with our expectations. The five optimal evaluation accuracies obtained were 96.24 %, 96.44 %, 96.51 %, 96.42 %, and 96.38 %. Based on this, the average reported accuracy of our proposed method is 96.40 %. The proximity of these individual values to the average further confirms the model's robustness. While we accept that additional trials may slightly influence the average performance, the effects are expected to be minimal.

Fig. 5 illustrates the smooth and effective convergence of both training and evaluation accuracies, highlighting the benefits of TL in enhancing performance and accelerating convergence. It also demonstrates that data augmentation and regularization effectively helped to prevent overfitting and accuracy fluctuations. Moreover, it shows that the dynamic learning rate improved accuracy after every five epochs. However, this effect is more pronounced in the earlier epochs, with diminishing returns in later stages.

Fig. 6 presents a normalized version of the confusion matrix, ensuring that varying data amounts across food categories do not bias our analysis. The matrix reveals that some specific categories (Noodle-Pasta, Rice, Soup, and Vegetable-Fruits) were classified with nearly perfect accuracy (around or above 99 %), while others (Dairy products, Dessert, and Egg) underperformed, with Dairy products showing the lowest performance.

While observing variations in classification performance across categories in ML and DL projects is typical, understanding these discrepancies is crucial for improvement. According to Fig. 6, we can discover that data imbalance did not heavily influence misclassification. Despite having the least data, "Rice" and "Noodles-Pasta" achieved outstanding performance, comparable to that of "Vegetable-Fruits" and significantly higher than that of "Desserts", which had the most significant proportion of data.

Moreover, "Noodles-Pasta" and "Dairy products" demonstrated starkly different outcomes, with one showing one of the highest accuracies and the other the lowest, despite having the same data volume. Notably, the mentioned items are just instances, and further insights can be drawn by examining and comparing Figs. 1 and 6 simultaneously.

The results support our hypothesis from Section 2.3 regarding dataset challenges due to similarities among categories. Food classes with similar colors, such as Bread and Dairy products, underperformed

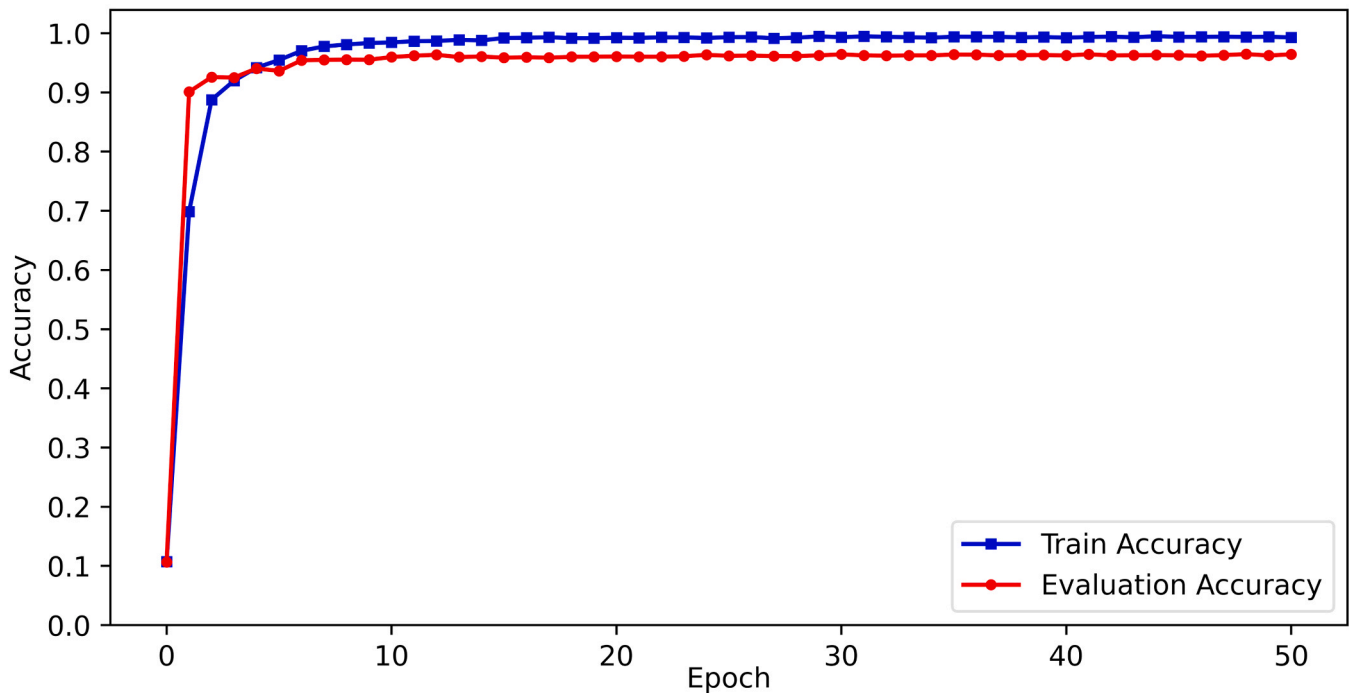


Fig. 5. Accuracy convergence for train and evaluation.

and exhibited mutual misclassifications, which will be further analyzed in Section 4.

In keeping with scientific rigor, it is essential to note that while the reported average accuracy of 96.40 % is based on five trials, Figs. 5 and 6 reflect the results of a single run. This, Fig. 5, and Fig. 6 may vary slightly between executions. Yet, it should be stated that despite potential differences in specific values, the relative rankings of classes, those with the best, average, and worst accuracies, remain relatively constant, supporting the reliability of our analysis and discussion.

Speed was also evaluated. Using a T4 GPU with 16 GB memory in Google Colab, training and evaluation speeds were approximately 1.01–1.03 batch/s and 1.95–2.01 batch/s, respectively. Hence, with batch sizes 16 for training and 32 for evaluation, the final speeds reached 16 and 62 images/second. Remarkably, even though speed can vary depending on batch size and hardware, the evaluation speed of 60 images/second on the T4 GPU, paired with high accuracy, is considered suitable for many real-world applications.

4. Discussion

4.1. Comparison with similar studies

This section examines research efforts that share our focus on accurate Food-11 image classification. In a study (Özsert Yiğit and Özyildirim, 2018), CaffeNet and AlexNet were used for food image recognition, initially obtaining accuracies of 80.51 % and 82.07 % with the SGD optimizer. Using Adam as the optimizer, the same models achieved enhanced accuracies of 83.7 % and 86.92 %. The images in their study had a resolution of 512×512 pixels, the highest among similar studies, leading to significantly more computations under the same circumstances.

(Islam et al., 2018a) conducted a study using a pre-trained InceptionV3 model paired with TL, data augmentation, and images resized to 299×299 pixels, resulting in an accuracy of 92.86 %. In (Islam et al., 2018b) a CNN was developed from the ground up. A pre-trained InceptionV3 model was incorporated into the research, with images resized to 224×224 pixels. After employing various DL techniques, the proposed approach achieved an accuracy of 74.7 %.

In (Suddul and Seguin, 2023), researchers initially designed a CNN from scratch, but this method resulted in a low accuracy of 64.6 %. Therefore, they used a pre-trained EfficientNetB2 model to improve their performance while using TL and data augmentation, ultimately reaching 94.5 % accuracy.

Leveraging the ensemble learning method, consisting of four pre-trained models named VGG19, ResNet50, MobileNet, and AlexNet, coupled with fine-tuning and data augmentation techniques, researchers (Bu et al., 2024) achieved 96.88 % accuracy, surpassing all previous studies on the Food-11 dataset.

In (Rokhva et al., 2024), the researcher used pre-trained MobileNetV2, full fine-tuning, and data augmentation to achieve 92.97 % accuracy with 256×256 images. They also studied other image sizes and achieved 88.2 %, 77.46 %, and 60.17 % accuracy with 128×128 , 64×64 , and 32×32 images. The evaluation speed for 256-sized images, which resulted in 92.97 % accuracy, was 291 images/second.

Given the outstanding performance of the EfficientNet family in Table 1, this study leveraged the pre-trained EfficientNetB7 as a backbone for feature extraction. The enriched feature map was enhanced with CBAM, an advanced attention module focusing on significant channels and spatial regions. After flattening the feature map and classifying it with 11 neurons in the FC layer, the model achieved around 96.40 % accuracy with 256×256 pixel images, surpassing previous single-model studies and approaching the 96.88 % accuracy obtained by (Bu et al., 2024), using an ensemble of four models.

The evaluation speed in this study reaches around 62 images per second on a T4 GPU. While this is considerably slower than the 291 images per second reported by (Rokhva et al., 2024) on the same hardware, the speed remains highly practical for most real-world applications. This is particularly true given that most kitchens are unlikely to serve over 60 dishes per second. A summary of the results from all studies on the Food-11 dataset can be found in Table 2.

The superior performance of this study among all single-model studies demonstrates the effectiveness of the proposed model, indicating that when the pre-trained EfficientNetB7 is fully fine-tuned and the extracted feature map enhanced with the CBAM attention module, the performance is higher than most prior research.

Last but not least, while this study was not as accurate as ensemble

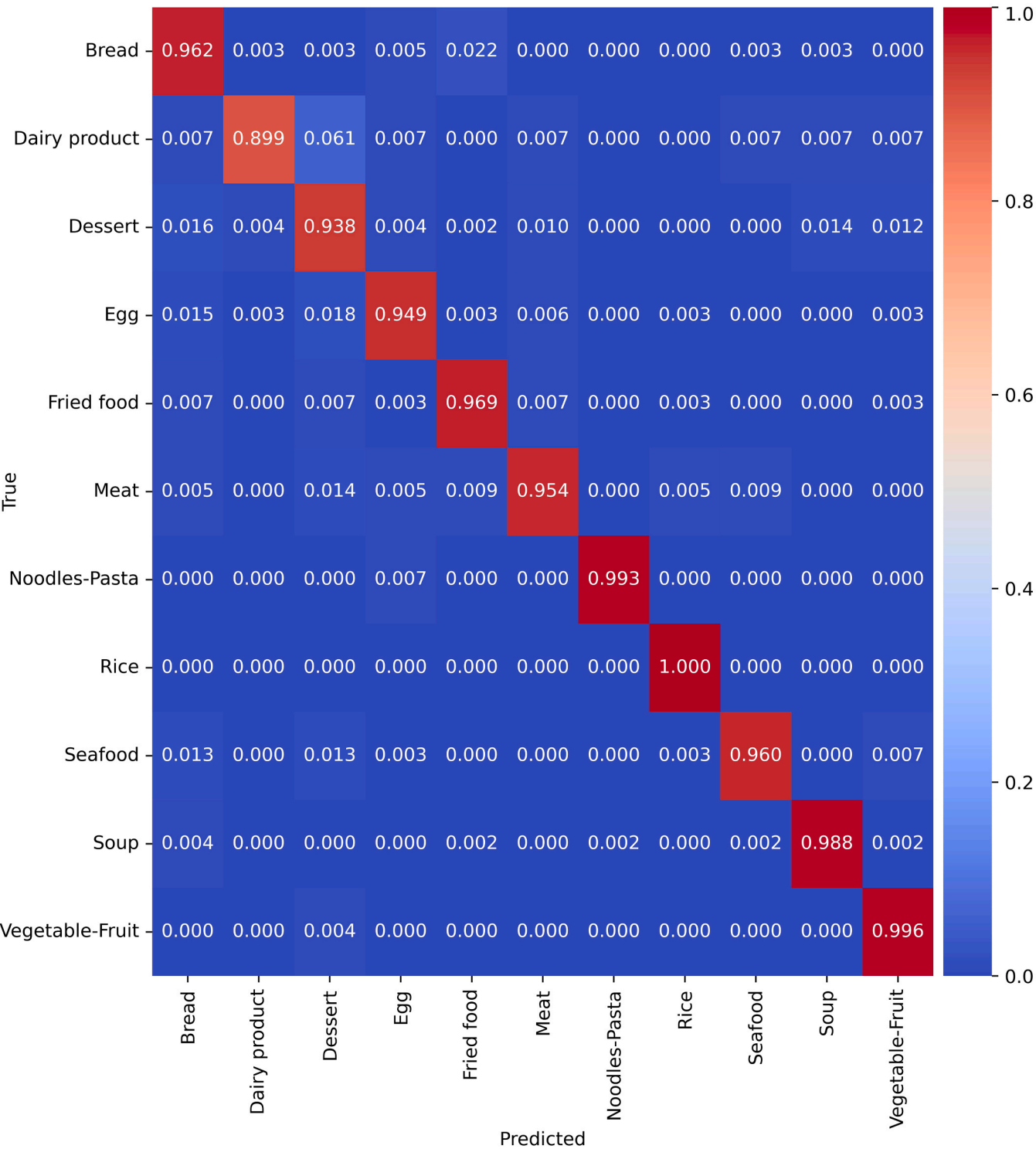


Fig. 6. Normalized confusion matrix.

methods (Bu et al., 2024), the current single model has fewer total parameters than the combination of four models. This reduction can be advantageous in inference mode, where backpropagation is absent, making the total parameter count and FLOPs critical considerations.

4.2. Studying the impact of employed technique

As shown in Fig. 5, TL proved to be effective for enhancing performance and accelerating convergence. Initially, at Epoch= 0, with no

pre-training, the accuracy was around 10 %. However, by using pre-trained weights and conducting full fine-tuning, training and evaluation accuracy surged to nearly 70 % and 90 %, after just one epoch (617 iterations with 9866 images, batch size 16). In addition, TL significantly reduced initial computational demands and allowed for smaller LR's at early stages, enabling the gradual application of even lower LR's later on, resulting in minimized accuracy fluctuations. Without TL, larger LR's and slower convergence with poorer performance within the same epochs would have been expected. TL also

Table 2
Comprehensive comparison with related studies.

Reference (time-based)	Employed DL model	Best reported accuracy	Additional information
(Özsert Yigit and Özyildirim, 2018)	CaffeNet & AlexNet (separately)	86.92 %	CaffeNet + SGD: 80.51 % CaffeNet + Adam: 83.7 % AlexNet + SGD: 82.07 % AlexNet + Adam: 86.92 % Image resolution: 512 × 512 (High)
(Islam et al., 2018a)	Inception-V3	92.86 %	TL + Data augmentation Image resolution: 299 × 299
(Islam et al., 2018b)	Classic CNN from the ground up	74.7 %	A combination of DL techniques Image resolution: 224 × 224
(Suddul and Seguin, 2023)	Classic CNN from the ground up & EfficientNetB2 (separately)	94.5 %	CNN from the scratch: 64.6 % EfficientNetB2: 94.5 % TL + Data augmentation
(Bu et al., 2024)	Ensemble Learning consisting four different models	96.88 %	Ensemble Learning consisting four models: VGG19, ReseNet50, MobileNet, and AlexNet. All models were pre-trained but fine-tuned. Weighted Ensemble Learning: 96.88 % Vote Ensemble Learning: 94.37 % <u>Best performance among all studies</u>
(Rokhva et al., 2024)	MobileNetV2	92.97 %	MobileNetV2 was pre-trained and fully tuned. Data augmentation was also used. Evaluation performance and speed: 256 × 256 size: 92.97 % - 291 image/s 128 × 128 size: 88.20 % - 452 image/s 64 × 64 size: 77.46 % - 626 image/s 32 × 32 size: 60.17 % - 702 image/s Hardware: T4 GPU
Current Study	EfficientNetB7 + CBAM + FC (tailored for 11 classes)	96.40 %	Pre-trained EfficientNetB7 (backbone) + Full Fine Tuning + CBAM (enhancement) + FC (tailored for 11 classes) TL + Diversified data + DL techniques. 96.40 % average accuracy and 62 image/s (for evaluation) Image resolution:

Table 2 (continued)

Reference (time-based)	Employed DL model	Best reported accuracy	Additional information
			256 × 256 <u>Best performance among single-model studies</u> <u>Surpassing the voting ensemble by more than 2 %</u> , and approaching the weighted ensemble (less than <u>0.5 %</u> difference)

eliminated the need to compute the mean and STD for Food-11, as the pre-calculated ImageNet values could be directly applied across all channels.

Regularization with a small L2 value (1e-5) proved to be highly effective, preventing overfitting and divergence. While evaluation accuracy fluctuated slightly around the optimal in later epochs, the model remained stable, eliminating the need for early stopping. Gradual LR halving every five epochs was particularly beneficial early on. An initial LR of 0.01 accelerated convergence but became less effective after partial network optimization, necessitating a reduced LR to improve convergence. Halving the LR led to a slight accuracy increase, enhancing convergence, though its effect diminished after 15–20 epochs.

Generalization through multiple executions aligned with our expectations and produced consistent results. The accuracy values ranged from 96.24 % to 96.51 %, closely matching the average of 96.40 %, validating the outcomes’ robustness and reliability.

4.3. Confusion matrix insights and rooms for improvement

Upon close examination of Figs. 6 and 1, it becomes clear that the underperformance of certain classes was not due to class imbalance. Instead, the intricate interplay of patterns and color shades posed significant challenges in distinguishing these classes. As anticipated in Section 2.3, classes such as Bread, Dairy Products, Dessert, Egg, and Fried Foods were expected to exhibit lower classification accuracy due to their frequent use of similar bright colors and patterns, making them susceptible to mutual misclassification. The results in Fig. 6 validate these expectations, showing that these underperforming categories had notable instances of mutual misclassification.

For example, hamburgers, pizzas, and bread filled with eggs or other ingredients were often classified as “Bread” but shared visual similarities with the “Fried Food” category. Interestingly, we expected “Bread” to be more misclassified as “Fried Food” than the reverse, which was validated by Fig. 6. Similarly, as expected, mutual inaccuracies were common among the Dessert, Dairy Products, and Egg classes due to their shared yellow and orange tones, visible in the confusion matrix.

This viewpoint can also justify the high performance of well-classified classes. “Rice” achieved high accuracy due to its unique granular texture and white color, while “Noodles-Pasta” stood out as the only filamentous class. The watery nature of “Soup” also facilitated accurate classification, and the distinctive colors in “Vegetable-Fruit” helped avoid confusion with other categories.

Several strategies can enhance classification performance, each holding significant promise. Firstly, increasing the granularity of certain classes may yield better results. For instance, splitting the broad “Fried Food” category into more specific types, such as French fries, fried chicken, and egg-fried bread, could lead to a more accurate classification. Even well-classified categories like “Seafood” and “Vegetable-Fruits” could benefit from finer subdivisions. By doing so, the deep networks can extract more object-specific and exclusive features, which are highly beneficial for accurate classification. Remarkably, detailing

classes means less number of data in each class. Therefore, given the data-hungry nature of the DL domain, increasing the data by collecting more samples or utilizing effective data augmentation becomes crucial.

Another promising strategy is hard example mining (HEM), which focuses on training the model with challenging samples that are difficult to classify, particularly those with similar textures and colors. A form of HEM can be done by modifying the loss function, such as Focal Loss, to prioritize complex cases. This would enable the model to refine its ability to distinguish subtle differences, such as the crispy texture that differentiates "Fried Food" from "Bread" or the nuances between creamy "Desserts" and "Dairy Products".

Incorporating techniques for model interpretability and explainability is crucial and could provide significant insights. For instance, Grad-CAM (Gradient-weighted Class Activation Mapping) can generate heatmaps that visualize which regions of an image influence the model's decisions, helping identify whether the model is focusing on relevant features or being misled by irrelevant elements. This method could be particularly useful in addressing the misclassifications between similar classes, such as "Bread" and "Fried Food."

Additionally, an approach involving the division of datasets and classes across multiple networks may prove effective. Training simpler classes on one network and more challenging ones on another can optimize models for specific tasks. Depending on the application, these networks could either share the same architecture or utilize different structures. This would also allow more computationally demanding techniques to be reserved for complex samples while simpler classes are handled efficiently by a less complex model.

Last but not least, combining these approaches can be highly effective. We guess a combination of some of these techniques (detailing classes, applying HEM and Focal Loss, and using different models for varying class difficulties) can offer a promising path for improving performance in our future work.

4.4. Studying the impact of hardware configuration

The training and evaluation speeds using T4 GPU and the specified batch in Section 2.8 sizes were approximately 16 and 62 images per second, respectively, with 62 being critical for real-world applications with the model's inference mode.

To conduct more research, the speed was also evaluated using the A100 GPU, which features 40 GB of GPU RAM, 83.5 GB of system RAM, and 235.7 GB of disk space. Under these conditions, the training and evaluation speeds reached approximately 3.80 and 8.04 batches per second, translating to 60.8 and 257.28 images per second.

Therefore, while the T4 GPU with 16 GB of memory could classify nearly 60 images per second in its inference model, the A100 GPU achieved an impressive rate of above 250 images per second. This indicates that the A100 GPU can be well-suited for more complex scenarios despite its higher cost. Nevertheless, we believe the T4 GPU remains adequate for many practical scenarios.

4.5. Policy implication

The application of advanced AI-driven techniques for food image classification has far-reaching implications in sectors like food safety, dietary management, food waste reduction, and consumer education (Lubura et al., 2022; Moumane et al., 2023). While no specific mobile app was developed, the proposed model demonstrates high accuracy and speed, making it suitable for diverse applications. This model can enhance food labeling and regulatory frameworks by ensuring compliance with nutritional standards and offering consumers reliable information (Chakraborty and Aithal, 2024; Gao et al., 2019). Additionally, its potential extends to AI-operated kitchens and smart refrigerators, optimizing processes and reducing waste. Policymakers can play a key role in integrating these AI-driven systems into food establishment monitoring frameworks (Ahmadzadeh et al., 2023).

Accurate food classification also supports dietary management, where healthcare providers can use it to track food intake, promote healthy eating habits, and address chronic diseases like obesity (Rehman et al., 2017). Furthermore, educational programs can leverage these AI advancements to promote awareness of food diversity, contributing to better public health outcomes and a more informed society (Shams et al., 2021).

5. Conclusion

Given the critical role of accurate and real-time food recognition in food production and consumption settings and addressing an existing research gap, in line with previous research, this study aimed to enhance AI-driven food classification performance while ensuring the necessary speed for real-time applications. We focused on the challenging Food11 dataset, consisting of 16643 food images across 11 diverse yet imbalanced classes, characterized by significant intra-class variations and inter-class similarities. From this dataset, 60 % was designated for training, while the remaining data was evenly split between validation and evaluation, which were used for hyperparameter tuning and final testing, respectively. The proposed approach combined the pre-trained state-of-the-art EfficientNetB7 architecture, recognized for its robust performance and cutting-edge compound scaling technique, with a Convolutional Block Attention Module (CBAM), which enhances both channel and spatial attention. By employing full fine-tuning of the EfficientNetB7 backbone alongside regularization techniques, dynamic learning rates, data augmentation, and refined hyperparameter tuning, the model achieved a remarkable average accuracy of 96.40 % on unseen evaluation data, with no signs of overfitting or performance inconsistencies. This performance surpassed all prior studies using single deep learning models and a voting ensemble learning approach, and also closely approached the 96.88 % accuracy achieved by the weighted ensemble method combining four pre-trained models with fine-tuning. Despite its complexity, the model maintained an inference speed of + 60 images per second using a budget-friendly T4 GPU, demonstrating its viability for real-world applications. Studying the confusion matrix also revealed that most misclassifications stemmed not from data imbalance or insufficiency but from visual similarities in color, pattern, and features across certain classes. Lastly, we comprehensively discussed the effectiveness of utilized techniques, the study's challenges and nuances, and identified the potential areas for future enhancements.

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CRedit authorship contribution statement

Shayan Rokhva: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Babak Teimourpour:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used "Perplexity.ai" in order to enhance language clarity and readability. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial

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